MODULE 3: CHAPTER 2.3 SYMMETRIC & ASYMMETRIC ENCRYPTION (TB PG 63–74)

Encryption transforms readable **Plain Text** into unintelligible **Cipher Text** using an algorithm and cryptographic key to protect confidentiality.

2.3.1 Symmetric Encryption (TB Pg 63–67)

- Uses the **same shared secret key** for both encryption and decryption.
- *Advantage:* Very fast and computationally efficient; ideal for encrypting large databases and hard drives.
- *Critical Vulnerability (Key Distribution Problem):* The sender must transmit the secret key to the receiver. If intercepted by an eavesdropper, all future communications are compromised.

2.3.2 Asymmetric Encryption (TB Pg 68–74)

Uses mathematically linked pairs of keys: a **Public Key** (openly published to the world) and a **Private Key** (kept strictly secret by the owner).

Operation	Key Used for Encryption	Key Used for Decryption / Purpose
Confidential Messaging	Recipient's **Public Key**	Recipient's **Private Key** (Only the recipient can open and read the message).
Digital Signature (Authenticity)	Sender's **Private Key**	Sender's **Public Key** (Proves message genuinely originated from sender and has not been tampered with).

2.3.3 Hashing & SSL/TLS Digital Certificates (TB Pg 70–74)

- **Hashing Function:** A one-way mathematical algorithm (e.g. SHA-256) that converts input data into a fixed-length hash digest. Impossible to reverse back into plain text. Used for storing passwords securely and verifying file integrity.
- **SSL / TLS Digital Certificates:** Issued by trusted Certificate Authorities (CAs). Verifies the authentic identity of a website, displays the padlock in browser URL, and exchanges asymmetric keys to establish secure encrypted symmetric sessions (HTTPS).

> 💡 **Exam Preparation Tip for Secondary 2 Students:**
> • Always state formulas clearly in file size calculations before writing numerical steps.
> • Remember that $1 ext{ KiB} = 1024 ext{ Bytes}$, whereas $1 ext{ KB} = 1000 ext{ Bytes}$. Cambridge penalizes confusing these prefixes!
> • For Parity Blocks, check both rows and columns: the intersection points directly to the erroneous bit.